Statistical Decision Making
45-752
Week 1, Session 1

Half of Any Solution Lies in Formulating the Problem

This file is named 01_slides.pptx

The accompanying Python notebook is 01_notebook.ipynb

Main goals of the course

- Provide understanding of key concepts in statistics
- Provide tools to undertake statistical analysis
- Help you become a better analytical manager (and leader!)

Policy on Taping and Recording

TEPPER

Taping and Recording of Classroom Activities

Classroom activities may be taped or recorded by a student for the personal use of that student, but may not be further copied, distributed, posted on the internet, published or otherwise used for any other purpose without the express written consent of Professor Dennis Epple. The same conditions apply for all documents and data sets provided in the course.

Instructor Information

TEPPER

Dennis Epple

Thomas Lord University Professor of Economics

Office: 4126

Email: epple@cmu.edu

- **Question and Answer Sessions:** Fridays at1:30pm. Sessions end at 3pm or when there are no more questions.
  - Tepper Room 2119-2120
  - I will permit access on Zoom.
  - Feel free to email in advance questions or topics that you would like me to cover.
  - Questions can be quite general. For example, you might ask me to review hypothes testing or review the plot command in Python.
- **Office Hours:** Friday 3:00pm to 4:00pm

Excellent TA!

Minhyuk Nam

(Head TA)

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- **Help Clinics:** Each Tuesday 6 - 8 PM in room to be announced.
- Office Hours by appointment.

Plan of Class Sessions

I will provide a pdf of PowerPoint slides in advance of each class session.

I will also provide a Python notebook with the commands used in the slides.

During the class session, we will work through the slides and the associated Python notebook.

For each statistical concept covered in the class, we will use data to apply the concepts and highlight managerial applications.

I think you will find these applications to quite valuable.

Approximate Schedule

Sessions 1-2: Review of Statistics Concepts; Statistical Quality Control; Review of single variable regression; Power of a Hypothesis Test; Designing an experiment;

Sessions 3-4: Multiple Variable regression, Testing Multiple Coefficients, Prediction, Introduction to Analysis of Time Series Data

Sessions 5-6: Indicator Variables, Alternative Functional Forms, Scaling of Coefficients ,

Sessions 7-8: Multicollinearity, Stepwise Regression, Testing functional form. Choice of Functional Form

Sessions 9-10: Modeling Choice, Logit and Probit

Sessions 11-12: Analysis of Variance, Analyzing Time Series Data

Session 13: Synthesis, Review

We may modify this schedule somewhat as we proceed.

Coursework

- You are assigned to a homework group.
- There will be 6 homework assignments that are to be turned in.
- There will be an in-class midterm exam.
- There will be a three-hour comprehensive final exam.

Grading

Course grade will be calculated as follows:

- Homework: 25%
- Midterm: 25%
- Participation (more next slide): 10%
- Final Exam: 40%

Your homework grade will be determined in part by assessments made by your peers:

- Each of you will be asked to do an assessment of contributions of other members of your group using an app called Group Member of Evaluation.

Discussion Questions and Class Participation

TEPPER

Warm not ***Cold*** Calling:

I will call on you on a Wednesday when you let me know you are in the mood to be called.

- If you want to participate, put a card with your name in an envelope that I will provide at the beginning of class. I will provide cards on which you can write your name.

I will randomly select from the those who submit names.

- If you are a lucky winner, I will ask "fairly easy" questions related to assignments, readings, or materials we have covered in class.
- If you aren't called, you still get participation credit.
- For full participation credit, you need to submit your name two times during the mini.
- There are five more Wednesdays, so don't wait too long before you start submitting your name.

In the past, students have said they like this approach.

Readings

In addition to slides and homework assignments, I will provide chapters of the book my son and I have written to help you learn the material.

You may find the class presentations and homework assignments to be sufficient to learn the material, but be sure to look through each chapter to check your understanding and fill any gaps.

The first chapter is a refresher on some basics of Colab and Python:

- 01_Introduction.pdf

Chapters 2 covers material in this slide deck. The chapter is titled:

- 02_Hypothesis_Testing_and_Statistical_Quality_Control.pdf

Some encouraging words…

If you keep up with the homework and readings and stay tuned in during class, you will succeed in this course.

You'll be well prepared for future Tepper Courses, and you will be well prepared to undertake statistical analysis in your work.

Analyzing data is genuinely interesting and I think you will enjoy it.

Python

Python has comprehensive capabilities.

We will use Google Colab, which is open-source. Hence it will be readily available to you in any job setting.

My son, Stephen and I have written a Python package named tprstats (pronounced Tepper stats) for your use.

This packages streamlines and greatly simplifies application of several statistical methods in Python.

Learning Statistics and Learning Python

In some respects, learning statistics is like climbing a ladder. You need to work your way up a step at a time.

There is, however, an important way in which learning statistics is different from climbing a ladder.

- With statistics, each step you take deepens your understanding of what came before.

The same comments apply to learning Python.

- You proceed a step at a time, and each new method you learn enhances your ability to use the methods that came before.

There is also synergy between learning statistics and learning Python.

- Learning more about one improves your ability use the other.

Artificial Intelligence

Carnegie Mellon has an arrangement with Microsoft for faculty and students to use Microsoft Copilot with prompts being kept private.

As you are learning Python, you may find Copilot to be useful in helping you refresh your memory about Python commands. Hence, I will provide some examples using Copilot in the course.

- You need to be aware that AI sometimes makes things up. Copilot may tell you to use a Python command that does not exist.
- Also, interpretation of statistical methods depends on the situation being studied. If you ask AI to interpret findings, it often gives a mechanical response that is not very informative.

The above comments are not to discourage you from using AI, but rather to highlight that you need to validate responses that you receive from AI.

Artificial Intelligence

It is important for you to learn to do statistical analysis without having to rely on Copilot or other AI tools.

In your job interviews, you may be asked questions about statistics. You obviously are not going to be in a position to use AI tools to answer questions during an interview.

Likewise, in future work when you are doing a presentation or participating in a discussion with your colleagues, you are not going to be in the position to need to turn to AI to try to answer questions.

Hence, unless announced otherwise, Copilot or other AI tools **may not** be used for the midterm exam and the final exam.

Questions?

Overview of Remainder of This Note Set

This note set contains a review of methods you learned in your probability and statistics course, including review of how these methods are implemented using Colab.

We will move relatively quickly through this review material.

The Python notebooks that we will use for the course and most of the data sets that we will use are in the Zip file posted on the course web page.

The following notebook has the Python commands for the slides that follow.

- 01_notebook.ipnyb

Overview of Remainder of This Note Set

The notebook named on the previous slide imports the following data sets. View the notebook to see where the commands are placed in the notebook.

Note the formatting of the commands. The data sets are in Excel files. To import a data file from the repository of Excel files you need to put a name of your choosing on the left-hand side. On the right-hand side, pandas.read_excel is used to read data files. As shown, you must enter in the parentheses “data/Data_File_Name.xlsx”

We will use these data files in our first two classes.

```
# import Data
Dollar_Bank = pandas.read_excel("data/Dollar_Bank.xlsx")
Dice_Data=pandas.read_excel("data/Dice_Data.xlsx")
Pgh_May_Temps = pandas.read_excel("data/Pgh_May_Temps.xlsx")
Theta_data = pandas.read_excel("data/Theta_data.xlsx")
Defects_data = pandas.read_excel("data/Defects_data.xlsx")
Diamonds_17_data = pandas.read_excel("data/Diamonds_17.xlsx")
```

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Click highlighted item to go to topic. Click to return to this slide.

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- Statistical Quality Control for a Binary Variable
- Summary of This Slide Deck
- Solutions to Practice Problems

Statistics Review

What is a Statistic?

A statistic is a number computed from a sample.

Example: The manager of the Dollar Bank in Squirrel Hill (a neighborhood near CMU) would like to know how far customers travel to get from their homes to the bank. A random sample of 30 people who have used Bank is drawn. This sample is from a list of people who have visited the bank in the past year. Google Maps is then used to find how far each person lives from the bank.

A statistic we might calculate from this sample is the sample mean $\bar{X}$.

- $\bar{X}$ is random.
- If we draw another random sample of 30, we will get a different value for $\bar{X}$.
- The distribution of possible values of $\bar{X}$ is the sampling distribution of $\bar{X}$.

Sampling Distribution

Sampling Distribution: The distribution of a statistic.

Example: Suppose mean driving distance for the population using the bank is μ and the population variance is σ^2.

Because we have a random sample, **regardless of the form of the population distribution:**

The expected value of $\bar{X}$ is μ i.e., E($\bar{X}$)=μ.

The population standard deviation of $\bar{X}$ is $\sigma/\sqrt{n}$.

- From this, we see that with a larger sample the standard deviation of $\bar{X}$ decreases (i.e., our estimate is more precise).

If population is normally distributed

- **If a population is normally distributed, then, for a sample of any size**, n, the distribution of $\bar{X}$ is *normally distributed* with population mean μ and std. dev. = $\sigma/\sqrt{n}$
- Of course, we generally do not know the distribution of the population; it may or may not be normal.

CENTRAL LIMIT THEOREM

- If sample size n is "large", then, regardless of the distribution of the population from which the sample is drawn, $\bar{X}$ is *normally distributed* with mean μ and std. dev. = $\sigma/\sqrt{n}$
- How large is large? That depends but for most well-behaved distributions, 30 is enough.
  - Well behaved means, for example, that the distribution is not highly skewed.
  - Larger samples are needed if the underlying population distribution is highly skewed.

Statistical Decision Making

Review of Hypothesis Testing

Testing a Hypothesis About A Population Mean

Hypothesis Testing Review

Two Key Concepts Underlying Hypothesis Testing Are:

Population: A population is the entire group we are interested in studying. For example, you might wish to study how advertising affects purchases by customers of your company, quality of vehicles produced by an automobile assembly line, satisfaction of company employees.

Often, the population is very large, and obtaining information for the entire population may be very costly or even infeasible. Hence, we analyze information from a sample.

Sample: A sample is a group of units selected from the population. We study the sample to try to answer questions we have regarding the population.

An Application

In our Dollar Bank example, customers of the bank are the population.

The manager selects a <u>simple random sample</u> of 30 customers.

- A simple random sample has the property that each individual in the population has an equal chance of being chosen on each draw.

The manager asks her assistant to use Google maps to find the travel time from home to the bank for each of the 30 customers.

The manager then calculates the average time, $\bar{X}$.

Central Limit Theorem

Recall that the Central Limit Theorem tells us the following:

If our sample size is large enough, we can treat $\bar{X}$ as a draw from a normal distribution with mean μ and standard deviation $\sigma/\sqrt{n}$.

If we knew σ we could do hypothesis tests using the normal distribution.

We do not know σ and must estimate σ.

This adds more uncertainty to our analysis.

We take account of this additional uncertainty using the t-distribution.

Estimating σ

We don't know the population variance, σ, so we estimate it from the sample.

We use s to denote the estimate of σ:

$$s = \sqrt{\frac{(X_i - \bar{X})^2}{(n-1)}}$$

We then divide s by $\sqrt{n}$. The result is the "Standard Error" of $\bar{X}$.

$$\text{std error} = \frac{s}{\sqrt{n}} = \frac{\sqrt{\frac{(X_i - \bar{X})^2}{(n-1)}}}{\sqrt{n}}$$

The standard error of $\bar{X}$ is a measure of the accuracy of our estimate of of $\bar{X}$.

Be sure you "get" the idea of a standard error. It is quite important.

Memorizing the formula is not important.

The Manager's Hypothesis

The manager thinks the average time it takes customers to travel to the bank is 15 minutes.

She wishes to investigate if the sample evidence supports her belief using a significance level α=.05.

The manager's null hypothesis, H_0: μ_0=15.

The alternative hypothesis: H_A: $\mu_0 \neq 15$

The actual population mean μ is unknown. The subscript zero is to indicate that μ_0 is the value the manager wishes to test.

We use the t-distribution to take account of the additional uncertainty that arises because we do not know σ and must estimate σ.

Illustration of t-Distribution with α=.05

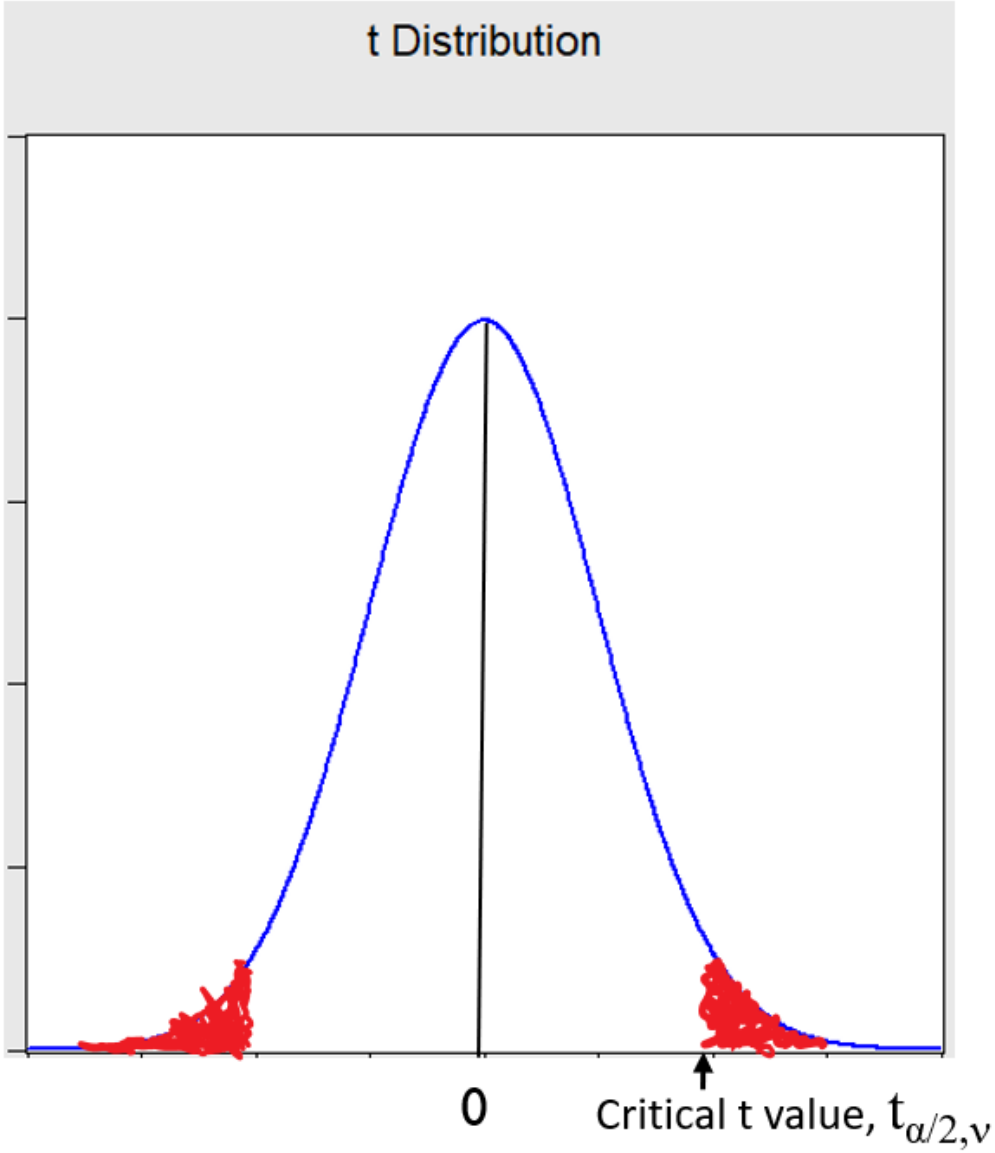
The t-distribution is centered on zero.

The critical t-value is denoted $t_{\alpha/2,v}$ where *v* is the degrees of freedom.

$\alpha/2$ of the area under the distribution lies to the right of $t_{\alpha/2,v}$, and $\alpha/2$ of the area lies to the left of $-t_{\alpha/2,v}$

Hence, with α=.05, 2.5% of area is under each tail, as shown in red.

Two-tailed tests are standard in practice. We will use only two-tailed tests.



Calculating Critical t-value

Our number of degrees of freedom is: ν= (n-1)=(30-1) = 29

The Python command to calculate the critical t-value is: t.ppf(1- α/2, ν)

In particular, with α=.05 and df=29, we issue the command: t.ppf(.975,29)

We obtain the critical t-value $t_{\alpha/2,\nu}$= 2.045

We can also obtain this value by asking Microsoft Copilot the following question: What is the t-statistic for a two-tailed test with significance level .05 and 29 degrees of freedom? Answer only for a two-tailed test.

Three ways to test a hypothesis

We reject the null hypothesis if:

1. The hypothesized null value, μ_0, is not in the confidence interval.
2. The calculated t-statistic is greater than the critical t value.
3. The p-value is less than α.

These all give the same decision.

The second and third approaches are more commonly used than the first, but it is instructive for you to also understand the first.

All three approaches are illustrated in the slides that follow.

First, the manager calculates the sample mean and sample standard deviation from the data for the sample of 30 customers and finds $\bar{X}$=13.9 and s=5.

Hence, the estimated standard error of $\bar{X}$ is $s/\sqrt{30} = 5/\sqrt{30} = .913$.

Method 1: Confidence Interval

In symbols, the confidence interval for μ is:

$$[\bar{X} - \mathrm{t}_{\alpha/2,\nu}\frac{s}{\sqrt{n}}, \bar{X} + \mathrm{t}_{\alpha/2,\nu}\frac{s}{\sqrt{n}}]$$

From previous slide, the manager used the data for the sample of 30 customers to calculate $\bar{X}$=13.9 and s=5.

Calculating the confidence interval, we obtain:

$$[13.9 - 2.045\frac{5}{\sqrt{30}}, 13.9 + 2.045\frac{5}{\sqrt{30}}] = [12.03, 15.77]$$

The manager's hypothesized null value μ_0=15 is in the above confidence interval. Hence, we do not reject her null hypothesis.

Method 2: Comparing Calculated t to critical t value

Using $\bar{X}$=13.9 and s=5, the manager then calculates the t-statistic:

$$t_{calc} = \frac{\overline{X} - \mu_0}{s / \sqrt{n}} = \frac{13.9 - 15}{5 / \sqrt{30}} = -\frac{1.1}{.913} = -1.205$$

This calculated t statistic is less in absolute value than the critical t value of 2.045.

Hence, the manager does not reject her null hypothesis that μ_0=15.

Method 3: Comparing p-value to α

From the previous slide, the manager's calculated t-statistic is -1.205

The manager calculates the two-tailed p-value with t=-1.205 and degrees of freedom=29.

Using Python: 2**t.cdf(-1.205,29)*

This tells us that the p-value is .24.

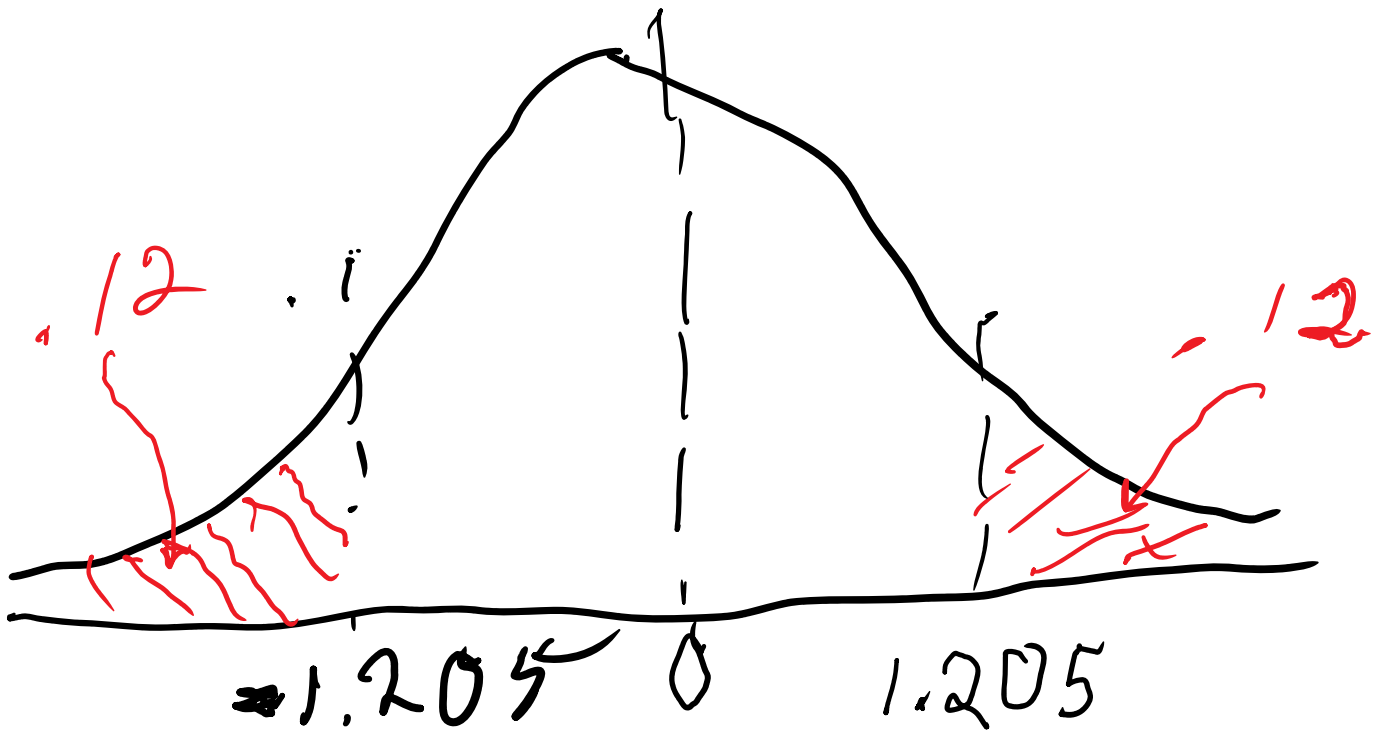
The p-value is greater than α so the manager does not reject her null hypothesis μ_0=15.

All three methods always yield the same result. Which approach you use is a matter of convenience.

Note: Always use the negative t-value in the above Python command, even if your calculated t-statistic is positive. I will explain the logic in class.

Illustration of the p-value

Recall that the calculated t-statistic is 1.205.



Interpreting the p-value

The p-value is increasingly used in hypothesis testing. Hence, it is important for you to be able to explain the p-value in words.

It is straightforward, but something of a mouthful.

In the experiment, the manager obtained a difference between her null hypothesis and the sample mean of 15-13.9=1.1.

We obtained a p-value of .24

Interpretation: If the null hypothesis is true, there is a 24% probability of obtaining a difference from the null as large or larger in absolute value than 1.1.

How could the manager choose her sample?

The manager wishes to choose a simple random sample of size 30 from the customers in the bank.

To choose a sample, the manager completed the following steps:

1. All customers' account numbers were put in Column A in an Excel spreadsheet.
2. In Cell B1, she entered a name of her choosing: MyRand
3. In cell B2 she entered: =RAND()
   This puts a random number in cell B2.
4. Next, by dragging down from cell B2, she put a random number in Column B for every customer.
5. She then sorted the two columns by column B.
6. After completing the sort, she chose the first 30 account numbers.

What do we mean when we say "by chance"

Suppose we have a fair coin.

- We flip the coin 10 times and obtain 6 heads.
- If it's a fair coin, why didn't we get 5 heads?
- The outcome of a sample of size 10 is random.
- The difference between the expected outcome (5 heads) and the sample outcome is due to chance.

Suppose the **population mean** distance from the bank is 15 miles.

- We draw a sample of size 30 and obtain a **sample mean** of 13.9.
- Why didn't we get a sample mean of 15?
- Our sample has people who, on average, are closer to the bank than the bank population.
- The difference between 15 and 13.9 is due to chance.

Statistical Decision Making

Understanding Quantiles

Understanding Quantiles

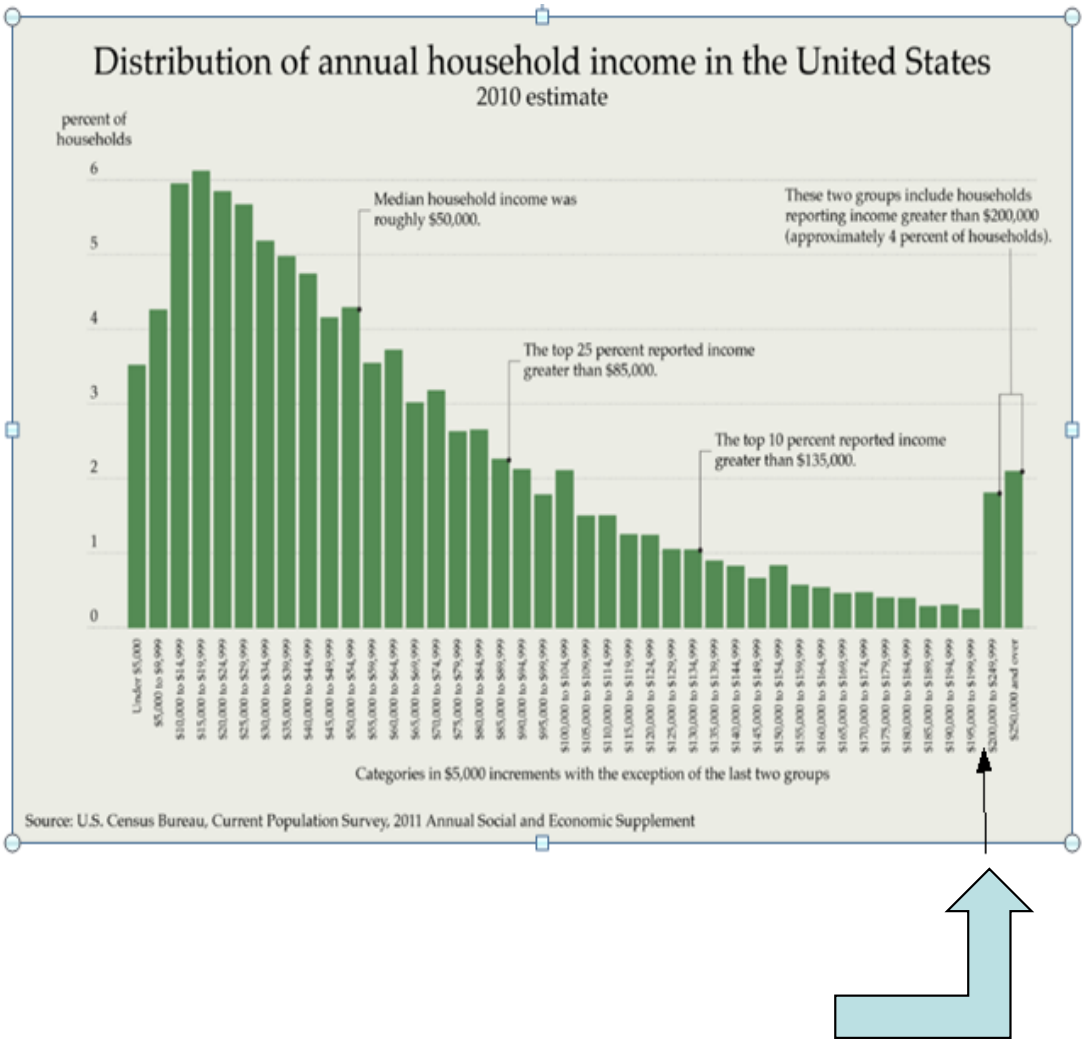
A quantile is the same thing as a percentile.

A quantile is a point on the horizontal axis of a distribution.

Example: In 2010, the 96th quantile of income in the US was $200,000 (see arrow);

- 96% of households had income less than $200,000;
- 96% of the area of the distribution was to the left of the point 200,000.

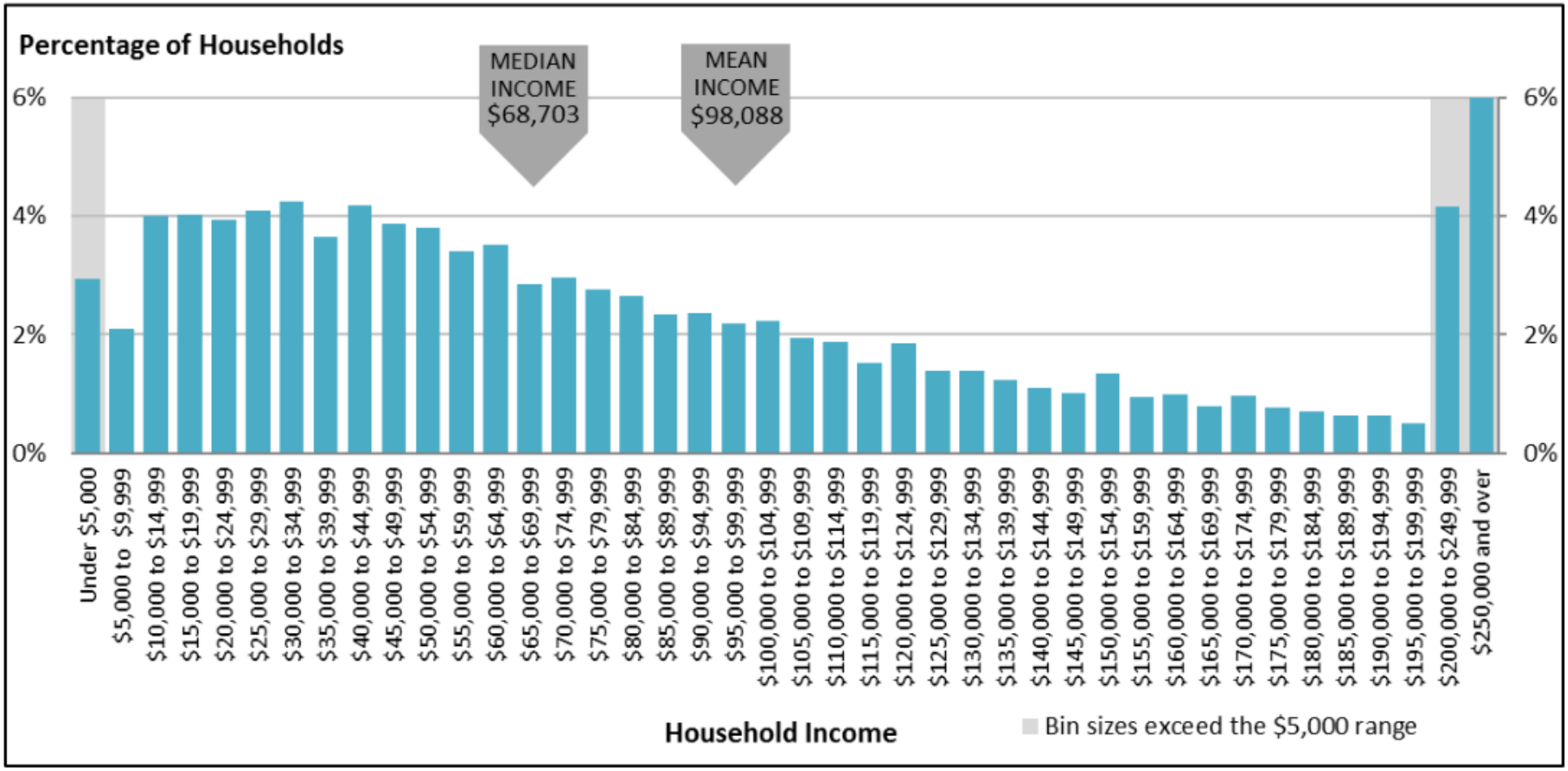
What was the 75th quantile?



US Household Income Distribution 2019

I thought you might be interested in seeing the income distribution for a more recent period. This is the most recent period for which the distribution data are available.

Figure 1. Distribution of Household Income, 2019



Source: https://sgp.fas.org/crs/misc/R44705.pdf

Calculating Quantiles from the Normal Distribution

The Normal Distribution:

- The standard normal distribution has a mean of 0 and standard deviation of 1. It is symmetrically distributed about its mean.
- We use $Z_{\alpha/2}$ to denote the (1-α/2) quantile of the standard normal.
- If α=.05, then the (1-α/2) quantile is $Z_{\alpha/2}$ =1.96.
- In words, the 97.5th quantile of the standard normal distribution is 1.96.
- Symmetry of the normal distribution implies that 95% of the area under the standard normal distribution lies between [-1.96,1.96]
- Hence, approximately 95% of the area under the normal distribution lies within two standard deviations of the mean.
- Using Python you can obtain quantiles of the standard normal using norm.ppf()
- For example to calculate the 97.5th quantile, issue the command:

  norm.ppf(.975)

Calculating Quantiles from the t-distribution

The t-Distribution:

- The 97.5th quantile of the **t-distribution** with 12 degrees of freedom is 2.18.
- Using Python, you can find this using: *t.ppf(.975,12)*
- Notice that the 97.5th quantile of the t-distribution with 12 d.f. is larger than 1.96 (the 97.5th quantile of the normal distribution).
- The lower the degrees of freedom, the greater the difference between the (1-α/2) quantile of the t and (1-α/2) quantile of the normal.
- The t-distribution is used when we do not know the variance of a distribution and must estimate the variance from our sample of data.

Calculating Quantiles From A Set of Data

To find a quantile of a variable named X, use the quantile() command.

quantile(X,.75)

To illustrate, import Dataset: Pgh_May_Temps

View the 75th quantile of Pittsburgh may Temperatures:

Pgh_May_Temps["Temperature"].quantile(0.75)

We obtain 67.875°. This tells us that, for 75% of days in May, Pittsburgh temperatures were 67.875° or below.

We can find multiple quantiles with one command. Below, we calculate the 10th, 50th, and 90th quantiles of Pittsburgh May temperatures:

Pgh_May_Temps["Temperature"].quantile([0.10, 0.5, 0.90])

We find that the median May temperature in Pittsburgh was 61.60°, and we see that 80% of May temperatures were between 51.20° and 72.97°.

10%	50%	90%
51.20	61.60	72.97

Summary

We have seen how to:

- Interpret quantiles.
- Calculate quantiles from theoretical distributions (normal, t).
- Calculate quantiles from empirical distributions (Pgh temperatures).

We will use quantiles frequently in this course.

The next slide provides practice questions followed by a slide with solutions to the questions.

Quantiles Practice

Using Python:

- Calculate the 16th quantile of the normal distribution.
- Calculate the 10th quantile of the t distribution with 20 df
- Find the area to the left of quantile -1 of the normal distribution.
- Find the area to the left of quantile -1 of the t distribution with 10 df.

Import the following data set:

Covid_Daily_Data_US_2020.xlsx

- Using a single line of Python code, calculate the 25th, 50th, and 75th quantiles of the variable in the file named New_Cases
- Interpret the 75th quantile.

Solutions are on the next slide.

Solution to Quantiles Practice

Full decimals are show below to make it easy for you to check your answer. In practice, we would round off to two or three digits after the decimal.

- Calculate the 16th quantile of the normal distribution: -0.9944579
- Calculate the 10th quantile of the t distribution with 20 df: -1.325341
- Find the area to the left of quantile -1 of the normal distribution. 0.1586553
- Find the area to the left of quantile -1 of the t distrib with 10 df. 0.1704466

Import the following data set and name it Covid_Data:
Covid_Daily_Data_US_2020.xlsx

- In the file there is a variable named New_Cases. Using a single line of Python code, calculate the 25th, 50th, and 75th quantiles of the variable in the file named New_Cases

You will obtain:

25%	50%	75%
21090	37918	61995

Interpret the 75th quantile. Here are two equivalent interpretations:

On 75% of days in 2020, the number of new Covid cases was 61,955 or lower. On 25% of days in 2020, the number of new Covid cases was 61,955 or higher.

Statistical Decision Making

Statistical Quality Control

Objectives of statistical quality control:

- Monitor production processes for quality
- Alert maintenance when intervention is needed

Control Problem for Theta & Theta Candy Company

The Theta & Theta Candy company produces a popular candy that comes in small pieces, each bearing the θ&θ company logo.

The company has machines to fill packages of θ&θ candy. If a machine is properly maintained, it will fill packages with an average of μ= 50 ounces of candy with accuracy given by a standard deviation of package weights of σ= .3 ounces.

The company wishes to monitor the production of candy packages:

- To evaluate whether the machines are producing properly.
- To alert maintenance if a machine is not producing according to specifications.

Measurement

The θ&θ company operates machines on a continuing basis.

- For each machine, it takes samples of 25 packages every 30 minutes and calculates the average weight of the 25 packages, $\bar{X}$
- If a machine is working properly, these sample means should have population mean μ=50 and standard deviation $\sigma_{\bar{X}} = {}^{.3}/_{\sqrt{25}}$.
- With samples of size 25, these sample means, $\bar{X}$, should be approximately normally distributed thanks to the central limit theorem.

Acceptance Interval

Let's plot a control chart with symmetric bounds for a 95% **acceptance interval**.

We wish to calculate bounds such that, if a machine is producing according to specifications, 95% of the sample mean values plotted on the chart will fall with the bounds on the control chart.

Find $Z_{\alpha/2}$ such that $\alpha/2$ of the area of the normal distribution lies to the right of $Z_{\alpha/2}$. With α=.05, $Z_{\alpha/2}$=1.96.

Then the control chart bounds are:

$$[\mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

Interpretation: (1-α) percent of the samples should have a sample mean within the above bounds if a machine is functioning according to specifications.

Let's Go to Python

For our application, we previously imported the file with our data: Theta_data.xlsx

The column named XBAR has data for 100 samples taken at 30 minute intervals.

The format of the command for the control chart is as follows. We enter the information for our chart in the command.

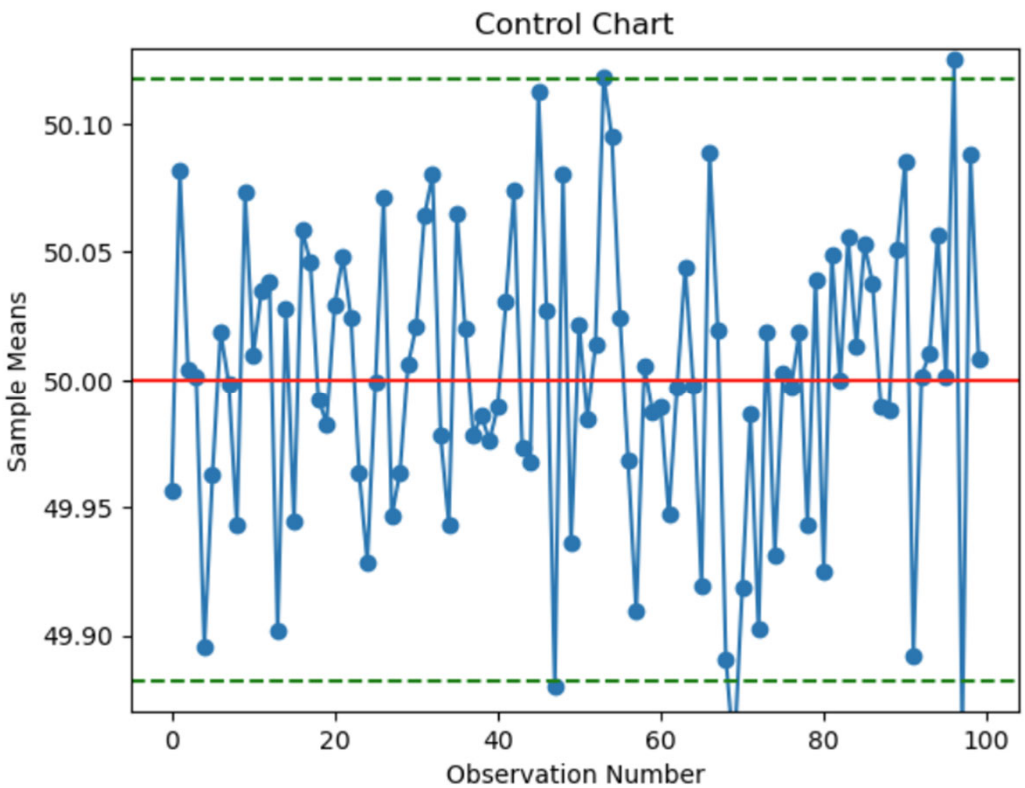
tprstats.control_chart(mu, sig, n, alpha, chartdata)

For our control chart, mu=50, sig=.3, n=25, alpha=.05.

We plot our control chart using the following:

```
# Following plots control chart for XBAR Theta & Theta Candy
tprstats.control_chart(mu=50, sig=0.3, n=25, alpha=0.05, data=Theta_data["XBAR"])
```

Control Chart for Machine 1



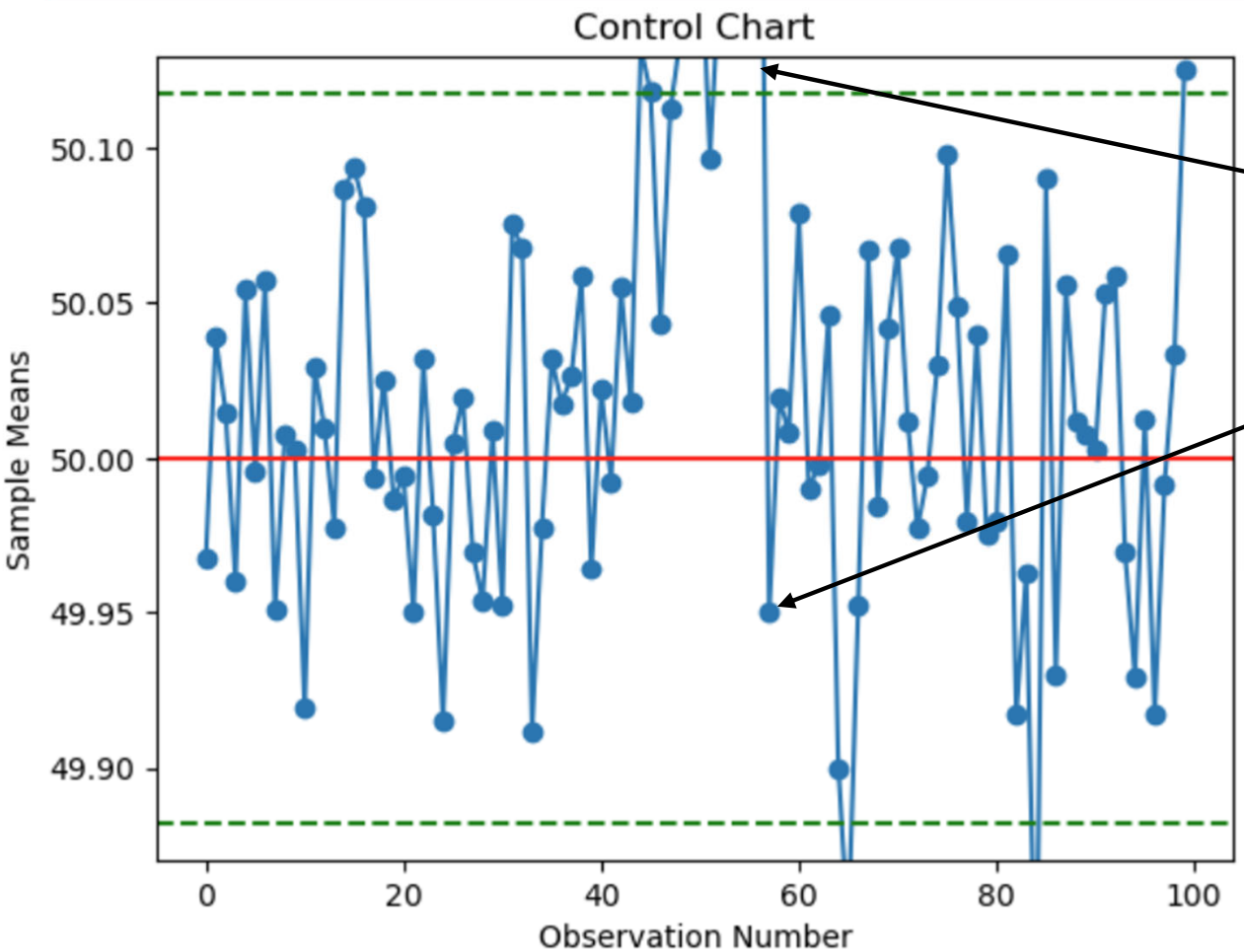
We see 5 of 100 observations fall outside of our confidence bounds.

We would expect on average to obtain 5 observations outside the bounds if the machine is operating at the accuracy specified by the manufacturer.

These results suggest that this machine performed quite well over this period of time.

Control Chart for Machine 2

```
#Following plots control chart for XBAR2 Theta & Theta Candy
# with alpha=.05
tprstats.control_chart(mu=50, sig=0.3, n=25, alpha=0.05, data = Theta_data['XBAR2'])
```



We see from this chart that the machine *did not* function properly for several periods.

Maintenance then corrected the problem. The correction was made before observation 58.

For the 43 observations from 58 onward, 3 are outside the bounds.

Thus, 3/43 observations, 7%, are outside the bounds. This is close to the 5% target, but future observations will reveal if the problem is fully resolved.

Quality Control Practice: Call Center Wait Times

Control charts can be used to monitor call center wait times.

Consider a call center, that provides user support for a company, Netflux, that operates a streaming service for tv shows and movies.

The wait time for each caller is automatically recorded. Excessive wait times can arise if call volume is high and/or the problems that callers are encountering take more than the usual amount of time to resolve.

The call Center alerts Netflux if there are persistent abnormal wait times that may reflect a technical problem with the Netflux streaming service. Expected mean customer wait time is μ=10 with st dev σ=15.

Earlier, we imported the following data set: Call_Center_Data.xlsx

- In this data set, variable WBAR contains the average wait times for 100 samples for calls to the Netflux call center.
- Each sample is the average wait time for n=30 successive callers.
- Using the above values of of μ and σ and n, Plot and evaluate the control chart for WBAR using α=.01.

Acceptance Interval vs Confidence Interval

Confidence Interval. We use a confidence interval when we do not know the population mean μ and standard deviation σ, and we wish to form an interval that contains the population mean with probability (1-α).

$$[\bar{X} - t_{\alpha/2,\nu}\frac{s}{\sqrt{n}}, \bar{X} + t_{\alpha/2,\nu}\frac{s}{\sqrt{n}}]$$

Acceptance Interval: We know the mean μ and standard deviation σ that characterize our production process, and we wish to form an interval that will contain a sample mean $\bar{X}$ with probability (1-α).

$$[\mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

Confidence intervals are used in a wide variety of applications. The use of acceptance intervals is more limited. Acceptance intervals are used primarily in statistical quality control and in design of experiments.

Recap

We saw how to construct a quality control chart.

We did an application to the θ&θ Candy Company.

We introduced the concept of an Acceptance Interval and compared to the concept of a Confidence Interval.

Our application thus far is to situations where we are monitoring outcomes with a continuous variable (ounces of candy per package, call center wait times.)

We next consider monitoring outcomes for a binary variable.

Statistical Decision Making

Statistical Quality Control
for Monitoring a Binary Outcome Variable

Example from a Tepper Summer Intern

Hello Professor Epple, how are you?

I'm working in the shipping operation of (major online retailer) this summer and part of my project is developing a tool that could send an alarm to the maintenance crew when a machine starts having too many defects.

These defects or incidents are not a measure of how defective they are (like a beer bottle filled at different levels), they are just a record that a package did not pass an inspection point (sensors on the line can tell if the package was not taped correctly or if the contents of the box are too high to close it). In which case the machine kicks those packages out and they require manual reprocessing.

So far the logic of the program is this: Over a predeDolld period of 10 minutes, the program gathers the number of packages processed by the machine and the number of defects it has found. For example 10 packages with bad tape out of 250 packages processed. I have an alarm for when that proportion is over 3% (an arbitrary number so far that was suggested by my manager). However, now I'm thinking a control chart is the way to go here to determine with some science when the number of defects are too many and the maintenance crew has to assist the operator.

Acceptance Interval

Recall our construction of an acceptance interval.

We have a machine that is to produce units of a product with a mean of μ and standard deviation σ.

Our acceptance interval is:

$$[\mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

Interpretation: (1-α) percent of the samples should have a sample mean within the above bounds if a machine is functioning according to specifications.

We used the normal distribution because we chose our sample size to be large enough that we can comfortably rely on the central limit theorem.

The same logic applies when we have a binary variable. The sample average for a binary variable (a sample proportion) will be approximately normally distributed if the sample size is sufficiently large to apply the Central Limit Theorem.

Control Chart for Binary Variable

Our summer intern used a control chart to monitor a binary outcome (package defective or not defective).

Let π be the probability that a package is defective if the machine is working properly. This is the mean of our binary variable, and is analogous to the population mean μ when we have a continuous variable.

The variance of a binary variable is π *(1- π). This is analogous to σ^2 when we have a continuous variable.

Hence, the standard deviation of the population is:

$$\sigma = \sqrt{\pi * (1 - \pi)}$$

The standard deviation of the average defect rate for of a sample of size n is then:

$$\frac{\sigma}{\sqrt{n}} = \sqrt{\frac{\pi * (1 - \pi)}{n}}$$

Summer Project Continued

The control chart limits for a binary variable are then:

$$[\pi - Z_{\alpha/2} * \sqrt{\frac{\pi * (1-\pi)}{n}}, \pi + Z_{\alpha/2} * \sqrt{\frac{\pi * (1-\pi)}{n}}]$$

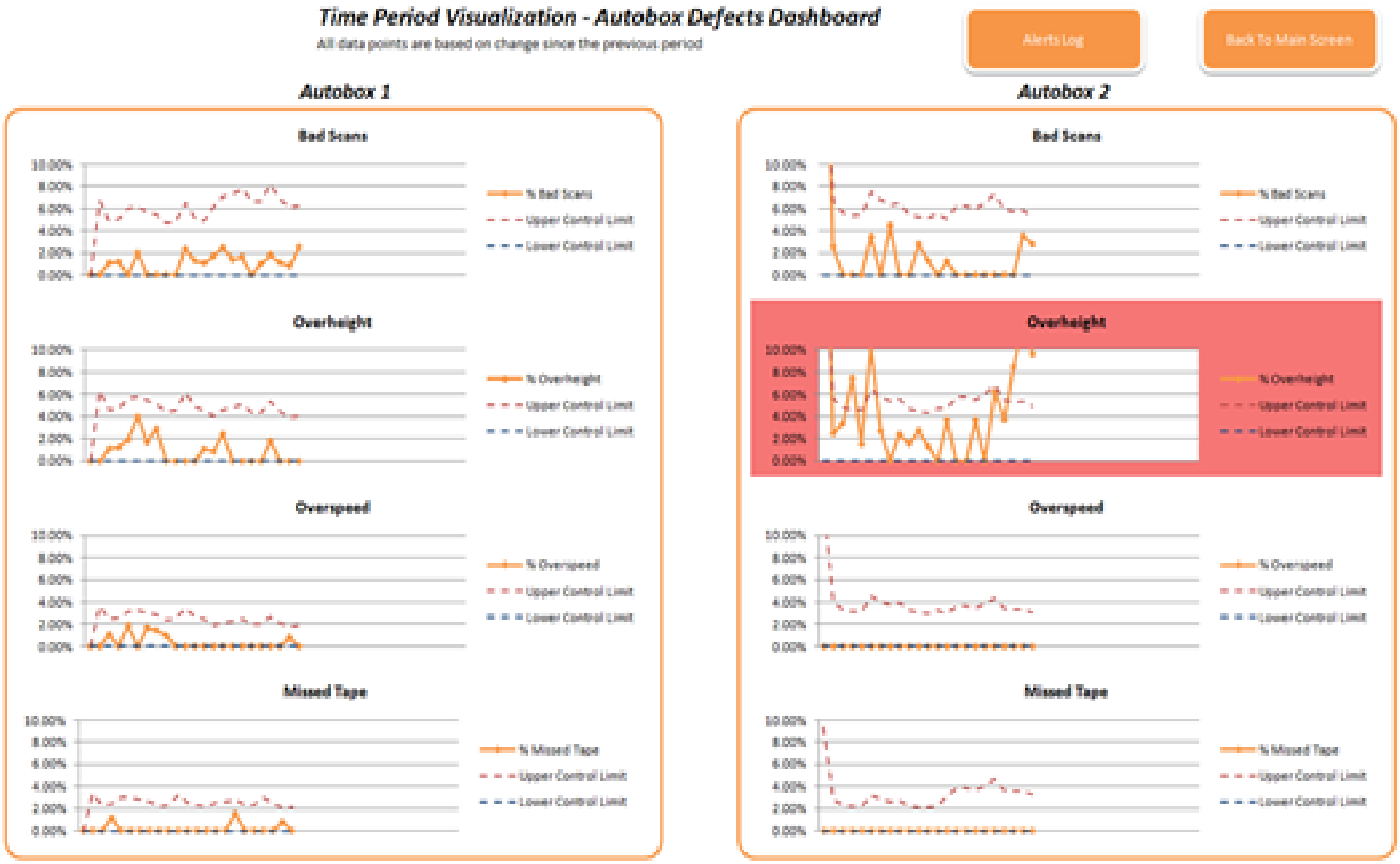
Our summer intern chose α=.003. Hence, $Z_{\alpha/2}$≈3.

Note: If defects are lower than expected, that is a good thing. Hence, no lower control limit is needed in this application.

Our intern set up charts to monitor four types of defects (bad scans, overheight, overspeed, missed tape).

The dashboard he designed in shown on the next slide. Note that the bound of each control chart varies because the number of packages, n, differs from one 5 minute interval to the next.

Control panel



Note the red border showing height out of control for Autobox 2.

Back to the summer project

He summarized as follows:

The system updated the control chart every 5 minutes with the sample average defect rate for packages produced during the five minute interval.

The control charts tool provides visibility to managers and operators and alerts them in case the defect rate for a 5 minute period is above the control limit on a tracking control chart.

This approach enables the team to identify and address issues as they occur and before they affect production output.

The result from this effort was very positive. Defect rates were reduced significantly and variation across shifts was also reduced.

This is an excellent example of the type of difference you can make by bringing statistical methods to bear in managerial decision making!

Binary Control Chart in Python

Application: A system produces an electrical device that must pass a series of tests. If the device fails the test, it must be discarded.

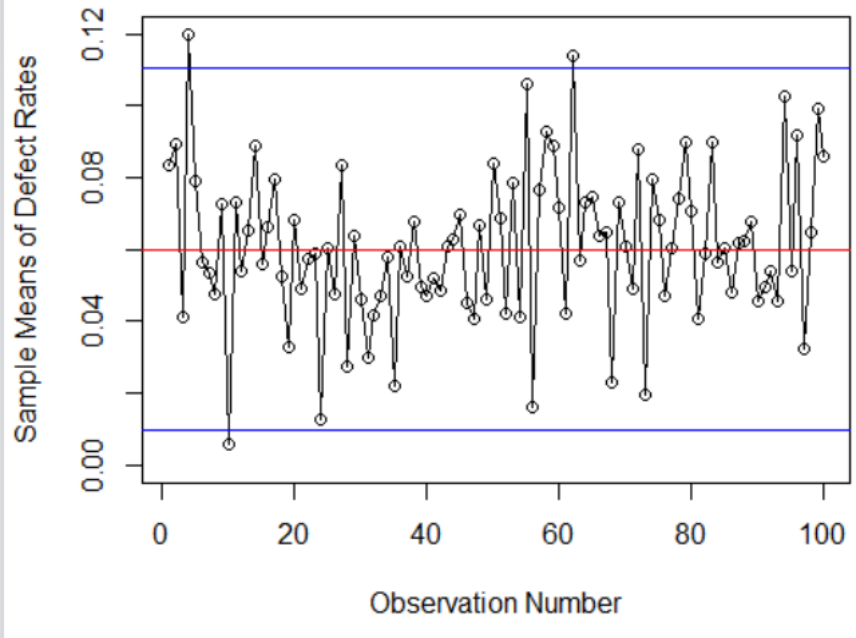
The mean defect rate when the process is working properly is .06.

We have a record of 100 samples with 85 observations per sample in Excel file Defects_data with variable named d_rate.

Import dataset Defects_data and issue the following command:

```
tprstats.control_chart_binary(0.06, 85, 0.05, Defects_data['d_rate'])
```

The control chart shows no problems over the period for which these 100 samples were taken.



Test Your Understanding:

Explain each of the following:

- Population mean: μ
- Population standard deviation: σ
- Standard deviation of $\bar{X}$: $\sigma/\sqrt{n}$

What is the relationship of each of the following to the above?

- Sample mean: $\bar{X}$
- Sample standard deviation: s
- Standard error of $\bar{X}$: $s/\sqrt{n}$

How are the above three items calculated?

Practice Question 1

You have a six-sided die.

- You wish to draw a random sample with five observations, n=5. How do you draw the sample?
  - You roll the die five times.
- Suppose you obtain the following: $X_1=2$, $X_2=4$, $X_3=3$, $X_4=1$, $X_5=5$.
- Calculate the following: $\bar{X}$, s^2, s, standard error of $\bar{X}$
- If the die is fair, the population mean $\mu=3.5$.
- We do not know if the die is fair. Suppose we wish to test the null hypothesis that the population mean for this die is 3.5. Is n=5 large enough to invoke the Central Limit Theorem to test this hypothesis?

Please work out your answers.

The calculations are easily done.

Then look at the next slide to check your solution.

Solution to Practice Question 1

You have a standard six-sided die.

- You draw a sample with five observations, n=5. How?
- You roll the die five times.
- You obtain the following: X_1=2, X_2=4, X_3=3, X_4=1, X_5=5.
- Calculate the following: $\bar{X}$, s^2, s, standard error of $\bar{X}$

Solution

$\bar{X} = 3$

$s^2 = 2.5$

$s = \sqrt{2.5} = 1.581$

$Standard\ Error\ of\ \bar{X} = s/\sqrt{5} = .7071$

n=5 is not nearly enough to invoke the Central Limit Theorem

Practice Question 2

- You draw a random sample with n=30 observations by rolling the die 30 times.
- Suppose you obtain $\bar{X} = 2.9$ and s^2=1.68.
- If the die is fair, the population mean μ=3.5.
- Test the null hypothesis μ_0=3.5.
- The alternative hypothesis is $\mu_0 \neq 3.5$.
- With n=30, we can invoke the Central Limit Theorem to test this hypothesis.
- Calculate the t-statistic and p-value. Use significance level .05.
- Please work out your answer.

Then check your solution on the next slide.

Solution to Practice Question 2

You draw a random sample with n=30 observations

- You obtain $\bar{X} = 2.9$ and s^2=2.5.
- Test the null hypothesis μ_0=3.5.
- Calculate the t-statistic and p-value. Use significance level α=05.

$Standard\ Error\ of\ \bar{X}$ =s/$\sqrt{30} = \sqrt{1.68}/\ \sqrt{30} = .2366$

$\mathrm{t} = (\bar{X} - 3.5)/.2366$=-2.536

Issue the Python command: 2*t.cdf(-2.536,29)

You obtain p-value=.0169

This is p-value is less than significance level α=05.

We reject the null hypothesis and conclude the die is not a fair die.

Question 3

When you have the data file, you can let Python do the work.

- I saved the data in variable X in the file: Dice_Data.xlsx
- Import the data file.

Then to test the null hypothesis that mu=3.5 enter and run the following command in Colab:

tprstats.test_population_mean(Dice_Data['X'],3.5)

You will obtain the following:

```
t_statistic                 -2.535983
p_value                      0.016859
confidence_interval_lower    2.416110
confidence_interval_upper    3.383890
sample_mean                  2.900000
```

This tells us: t = -2.536, p-value = 0.0169

These are the values we calculated on the previous slide.

Test Your Understanding

What is the conceptual difference between μ and μ_0?

Where was the standard error of $\bar{X}$ used in the hypothesis test?

Where was μ_0 used in the hypothesis test?

If we measured distance in kilometers instead of miles in the Dollar Bank data, would $\bar{X}$ change?

If we measured distance in kilometers instead of miles, would μ_0 change?

If we measured distance in kilometers instead of miles, would the standard error of $\bar{X}$ change?

If we measured distance in kilometers instead of miles, would the t-statistic or p-value change?

Please try to answer the above questions. Then go to the next slide to check your answers.

Answers To "Test Your Understanding" Questions

Q. What is the conceptual difference between μ and μ_0?

A. μ is the true population mean which is unknown to us, μ_0 is our hypothesized value of the mean

Q. Where was the standard error of $\bar{X}$ used in the hypothesis test?

A. The standard error of $\bar{X}$ is the denominator of the t-statistic

Q. Where was μ_0 used in the hypothesis test?

A. The numerator of the t-statistic is the difference between the sample mean and μ_0

Q. If we use distance in kilometers instead of miles, would $\bar{X}$ change?

A. Yes. There are 1.61 kilometers per mile. Hence, to convert from miles to kilometers, we would multiply $\bar{X}$ by 1.61.

Q. If we measure distance in kilometers instead of miles, would the standard error of $\bar{X}$ change?

A. Yes. The standard error of $\bar{X}$ measured in kilometers would 1.61 times the standard error of $\bar{X}$ measured in miles. You can confirm this by checking the formula for the standard error of of $\bar{X}$.

Q. If we measure distance in kilometers instead of miles, would μ_0 change?

A. Yes, instead of 15 miles, the null hypothesis would be 1.61*15=24.15 kilometers.

Q. If we measured distance in kilometers instead of miles, would the t-statistic or p-value change?

A. No. The values in both the numerator and denominator of the t-statistic would be multiplied by 1.61, so the multiplier 1.61 would cancel.

Q. Why is the above important?

A. It tells us that the results of our hypothesis test don't depend on the units of measurement of our variables. If we don't reject the null hypothesis of 15 miles, we wouldn't reject the null hypothesis of 24.15 kilometers.

p-value Recap (for your review)

Loosely, you can think of the p-value as the probability that the null hypothesis is true.

- If the p-value is small, the null hypothesis is probably false.

In more precise terms, the p-value is the probability of getting a difference between the estimated value and null value as large or larger than the value we actually obtained if the null hypothesis is true.

- In our Dollar Bank example, the difference between the estimate and the null value is 1.1.
- We calculated that an estimate this far from the null would occur by chance with a probability (p-value)= .24.
- We would reject the null only if the difference between the estimate and the null would occur by chance with a probability less than the significance level, α, which is .05 (i.e., if p-value < α).
- Our p-value is greater than α (i.e., .24 > .05) so we do not reject the null.

Recap

In these slides:

We saw how to construct a quality control chart to monitor a binary variable.

We reviewed an application to the shipping department of a major online retailer.

We did a practice application to monitoring defect rates for an electrical device.